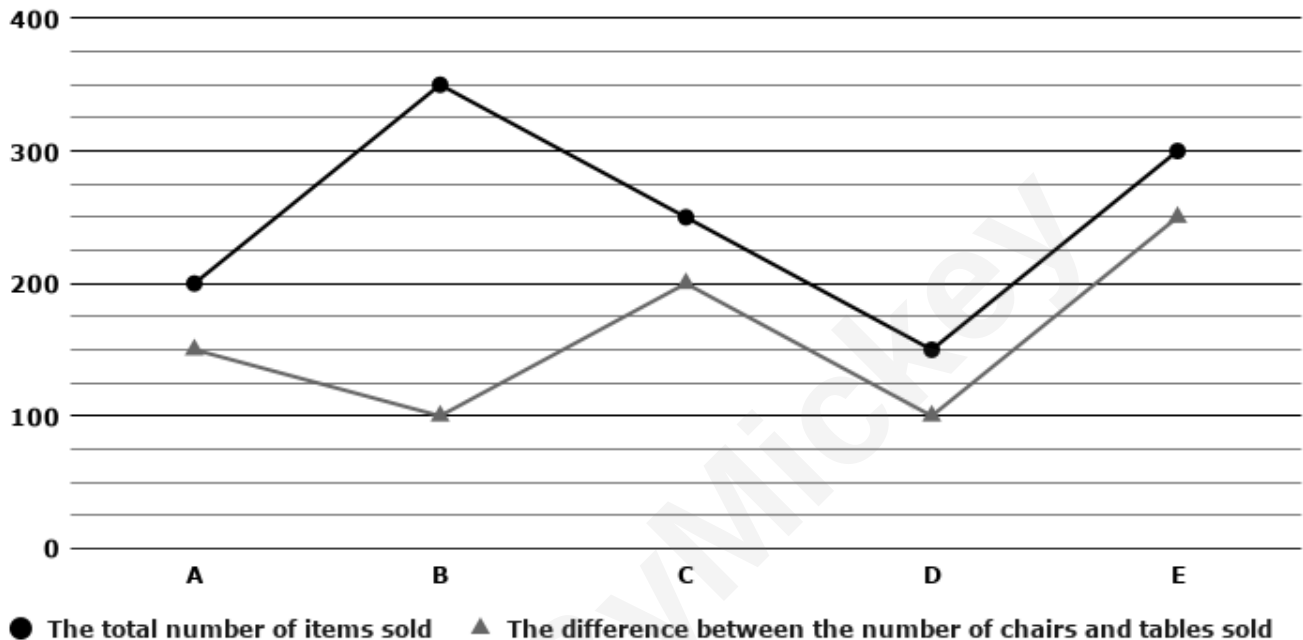


1. Questions

Study the following information carefully and answer the questions given below.

The line graph given below shows the total number of chairs and tables sold by five different shops namely A, B, C, D and E along with the difference between the number of chairs sold and the number of tables sold by each shop.



Notes:

(i) The total number of items sold = The number of chairs sold + The number of tables sold.

(ii) In each shop the number of chairs sold is more than the number of tables sold.

The number of chairs sold by shop D is $x\%$ of the total number of items sold by shop C and the number of tables sold by shop C is $y\%$ of the total number of items sold by shop A, find the value of $(x + y)$.

- 42.5
- 61.5
- 35.5
- 62.5
- 58.5

2. Questions

Find the ratio between $(\frac{3}{5})$ th of the number of chairs sold by shop A and 33.33% of the total number of items sold by shop E.

- 19 : 18

- b. 21 : 20
- c. 17 : 16
- d. 20 : 19
- e. 15 : 14

3. Questions

The total number of tables sold by shops B and E together is what percentage of the total number of items sold by shop D?

- a. 120%
- b. 90%
- c. 100%
- d. 80%
- e. 110%

4. Questions

Find the difference between the total number of chairs sold by shops A, C and E together and the total number of tables sold by shops B, C and D together.

- a. 500
- b. 400
- c. 300
- d. 600
- e. 700

5. Questions

If the average of the total number of items sold by shops B and C is $3p$, find the value of p .

- a. 110
- b. 80
- c. 90
- d. 120
- e. 100

6. Questions

What is the difference between the number of chairs sold by shop B and shop D?

- a. 100

- b. 120
- c. 110
- d. 130
- e. 150

7. Questions

Study the following information carefully and answer the questions given below.

The area of a rectangular field is 500 m^2 and its length is 25 m. The perimeter of a square field is equal to two-third of the perimeter of the rectangular field where the side of the square field is 'x' m.

If the cost of planting grass in the square field is Rs. 12 per square meter, what is the total cost to cover the entire square field with grass?

- a. Rs. 2700
- b. Rs. 2400
- c. Rs. 2500
- d. Rs. 2800
- e. Rs. 2600

8. Questions

Find the difference between the area of the rectangular field and the area of the square field.

- a. 245 m^2
- b. 235 m^2
- c. 255 m^2
- d. 225 m^2
- e. 275 m^2

9. Questions

Study the following information carefully and answer the questions given below.

The table given below shows the total number of cakes (vanilla and chocolate) sold by five different shops P, Q, R, S and T along with the percentage of chocolate cakes sold by each shop.

Shops	The total number of cakes sold (vanilla + chocolate)	The percentage of chocolate cakes sold
P	250	40%
Q	375	66.66%
R	128	75%
S	320	55%
T	200	67%

Note: The total number of cakes sold = The number of vanilla cakes sold + The number of chocolate cakes sold.

If Shop U sold 20% more total cakes than Shop S, and the ratio of vanilla cakes to chocolate cakes sold by Shop U is 3 : 5, find the number of chocolate cakes sold by Shop U.

- a. 220
- b. 230
- c. 210
- d. 240
- e. 250

10. Questions

Find the difference between the total number of vanilla cakes sold by Shop R and Shop T together and the number of chocolate cakes sold by Shop S.

- a. 78
- b. 63
- c. 85
- d. 72
- e. 81

11. Questions

The number of chocolate cakes sold by Shop P is what percentage less than the number of vanilla cakes sold by Shop Q?

- a. 10%

- b. 15%
- c. 20%
- d. 25%
- e. 30%

12. Questions

Find the average of the number of chocolate cakes sold by Shop P, Shop R, and Shop T.

- a. 130
- b. 80
- c. 120
- d. 90
- e. 110

13. Questions

Find the ratio of the number of vanilla cakes sold by Shop P to the number of vanilla cakes sold by Shop S.

- a. 18 : 19
- b. 25 : 24
- c. 21 : 22
- d. 23 : 25
- e. 17 : 16

14. Questions

The ratio of milk and water in a 60 liters mixture is 5 : 1 respectively. 30 liters of the mixture are taken out and y liters of water are added. If the ratio of milk to water in the final mixture is 5 : 6 respectively, find the value of y.

- a. 25
- b. 20
- c. 15
- d. 10
- e. 35

15. Questions

The ratio of the present ages of P and Q is 5 : 4. After 3 years, the ratio of their ages will become 11 : 9. If R is 8 years younger than P, then find the sum of the present ages of Q and R.

- a. 38 years
- b. 40 years
- c. 46 years
- d. 42 years
- e. 36 years

16. Questions

A and B started a business by investing Rs. 15000 and Rs. 24000 respectively. After 4 months, C joined them with an investment of Rs. 18000. If the total profit earned at the end of the year is Rs. 68000, what is C's share in the profit?

- a. Rs. 12000
- b. Rs. 18000
- c. Rs. 16000
- d. Rs. 24000
- e. Rs. 21000

17. Questions

The ratio of the speed of boat A to the speed of boat B in still water is 4 : 5. The speed of the stream is 6 km/hr. Boat A takes 5 hours to travel 150 km downstream. Find the time taken by boat B to travel 96 km upstream.

- a. 2 hours
- b. 5 hours
- c. 3 hours
- d. 4 hours
- e. 6 hours

18. Questions

A shopkeeper marks an article at 60% above its cost price and earns a profit of Rs. 800 by giving a 25% discount on the marked price. If he sells the article at a 15% discount on the marked price, find the profit earned by the shopkeeper.

- a. Rs. 1250
- b. Rs. 1490
- c. Rs. 1530
- d. Rs. 1320
- e. Rs. 1440

19. Questions

A is 50% more efficient than B. A alone can complete a piece of work in 48 days. A and B start working together but both leave after 12 days of work. The remaining work is completed by C alone in 6 days. In how many days can B and C working together complete the entire work?

- a. 5 days
- b. 6 days
- c. 8 days
- d. 9 days
- e. 7 days

20. Questions

Samay invested Rs. 18000 in a scheme offering simple interest at a rate of $r\%$ p.a. for 3 years and received an interest of Rs. 24300. Find the compound interest (compounded annually) on the same sum for 2 years at a rate of $(r + 5)\%$ p.a.

- a. Rs. 23500
- b. Rs. 21400
- c. Rs. 15400
- d. Rs. 28000
- e. Rs. 22500

21. Questions

What value should come in the place of (?) in the following questions.

$$5/9 * 729 + 345 - ? * 3 = 15^2$$

- a. 134
- b. 129
- c. 104
- d. 175
- e. 158

22. Questions

$$\sqrt{6400} * 5 \div 8 = ?$$

- a. 50
- b. 30
- c. 40

d. 20

e. 60

23. Questions

$$845 \div 5 - 25 * 4 + 41 * 5 = ?$$

a. 247

b. 321

c. 259

d. 324

e. 274

24. Questions

$$(? + 72 \div 12) * 30 = 240$$

a. 2

b. 3

c. 4

d. 1

e. 5

25. Questions

$$(8)^3 * (16)^2 \div (32)^2 = (2)^?$$

a. 5

b. 4

c. 7

d. 6

e. 8

26. Questions

What approximate value will come in the place of (?) in the following questions?

$$(5.02/7.98) \text{ of } 319.95 + 19.99\% \text{ of } 749.98 = ? * 4.99$$

a. 70

b. 60

c. 50

d. 80

e. 90

27. Questions

$$32.02 * 14.99 - 18.01 * 11.95 + 14.99\% \text{ of } 600.15 = ?$$

a. 412

b. 354

c. 427

d. 395

e. 482

28. Questions

$$(215.98)^{1/3} * (728.95)^{1/3} + 14.99 * 5.95 = ?^2$$

a. 44

b. 4

c. 12

d. 144

e. 22

29. Questions

$$59.95\% \text{ of } 1250.23 + \sqrt{4760.98} * 3.99 = ? + 145.99$$

a. 793

b. 821

c. 700

d. 880

e. 664

30. Questions

$$119.98 \div 4.95 * 15.01 - 25.02\% \text{ of } 439.98 = ?$$

a. 200

b. 100

c. 350

d. 150

e. 250

31. Questions

What value will come in the place of (?) in the following number series?

6, 4, ?, 11, 39, 189

- a. 7
- b. 6
- c. 3
- d. 5
- e. 4

32. Questions

46, 59, 76, 95, 118, ?

- a. 138
- b. 123
- c. 147
- d. 125
- e. 134

33. Questions

?, 8, 12, 39, 55, 180

- a. 5
- b. 7
- c. 4
- d. 9
- e. 2

34. Questions

16, 9, 10, ?, 33, 83.5

- a. 16
- b. 12
- c. 15
- d. 13

e. 17

35. Questions

13, 18, 25, 34, ?, 58

- a. 49
- b. 37
- c. 42
- d. 39
- e. 45

36. Questions

Find the wrong number in the following number series.

10, 12, 26, 81, 328, 1645

- a. 10
- b. 26
- c. 1645
- d. 328
- e. 12

37. Questions

9, 22, 48, 87, 140, 204

- a. 22
- b. 140
- c. 204
- d. 48
- e. 87

38. Questions

18, 34, 70, 134, 234, 380

- a. 70
- b. 18
- c. 380
- d. 134

e. 234

39. Questions

21, 29, 41, 56, 77, 106

- a. 106
- b. 41
- c. 21
- d. 29
- e. 56

40. Questions

33, 62, 93, 130, 171, 215

- a. 62
- b. 33
- c. 130
- d. 171
- e. 215

41. Questions

The following questions contain two equations as I and II. You have to solve both equations and determine the relationship between them and give answer as,

I). $x^2 - 28x + 195 = 0$

II). $y^2 - 30y + 224 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established
- d. $x < y$
- e. $x \leq y$

42. Questions

I). $x^2 - 18x + 77 = 0$

II). $y^2 - 24y + 143 = 0$

- a. $x > y$

- b. $x \geq y$
- c. $x = y$ or relationship cannot be established
- d. $x < y$
- e. $x \leq y$

43. Questions

I). $x^2 + 17x + 72 = 0$

II). $y^2 + 21y + 110 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established
- d. $x < y$
- e. $x \leq y$

44. Questions

I). $x^2 + 19x + 88 = 0$

II). $y^2 + 11y + 30 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established
- d. $x < y$
- e. $x \leq y$

45. Questions

I). $x^2 - 12x + 35 = 0$

II). $y^2 - 9y + 20 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established
- d. $x < y$
- e. $x \leq y$

Explanations:

1. Questions

Shop A:

The total number of items sold by shop A = 200

The number of chairs sold by shop A = $(200 + 150)/2 = 175$

The number of tables sold by shop A = $200 - 175 = 25$

Shop B:

The total number of items sold by shop B = 350

The number of chairs sold by shop B = $(350 + 100)/2 = 225$

The number of tables sold by shop B = $350 - 225 = 125$

Shop C:

The total number of items sold by shop C = 250

The number of chairs sold by shop C = $(250 + 200)/2 = 225$

The number of tables sold by shop C = $250 - 225 = 25$

Shop D:

The total number of items sold by shop D = 150

The number of chairs sold by shop D = $(150 + 100)/2 = 125$

The number of tables sold by shop D = $150 - 125 = 25$

Shop E:

The total number of items sold by shop E = 300

The number of chairs sold by shop E = $(300 + 250)/2 = 275$

The number of tables sold by shop E = $300 - 275 = 25$

Shops	The number of chairs sold	The number of tables sold	The total number of items sold
A	175	25	200
B	225	125	350
C	225	25	250
D	125	25	150
E	275	25	300

Answer: D

The number of chairs sold by shop D = 125

The total number of items sold by shop C = 250

So,

$$x\% = 125 / 250 * 100 = 50\%$$

The number of tables sold by shop C = 25

The total number of items sold by shop A = 200

So,

$$y\% = 25 / 200 * 100 = 12.5\%$$

The required answer = $(50 + 12.5) = 50 + 12.5 = 62.5$

Hence, the correct answer is option D.

2. Questions

Shop A:

The total number of items sold by shop A = 200

The number of chairs sold by shop A = $(200 + 150)/2 = 175$

The number of tables sold by shop A = $200 - 175 = 25$

Shop B:

The total number of items sold by shop B = 350

The number of chairs sold by shop B = $(350 + 100)/2 = 225$

The number of tables sold by shop B = $350 - 225 = 125$

Shop C:

The total number of items sold by shop C = 250

The number of chairs sold by shop C = $(250 + 200)/2 = 225$

The number of tables sold by shop C = $250 - 225 = 25$

Shop D:

The total number of items sold by shop D = 150

The number of chairs sold by shop D = $(150 + 100)/2 = 125$

The number of tables sold by shop D = $150 - 125 = 25$

Shop E:

The total number of items sold by shop E = 300

The number of chairs sold by shop E = $(300 + 250)/2 = 275$

The number of tables sold by shop E = $300 - 275 = 25$

Shops	The number of chairs sold	The number of tables sold	The total number of items sold
A	175	25	200
B	225	125	350
C	225	25	250
D	125	25	150
E	275	25	300

Answer: B

$3/5$ th of the number of chairs sold by shop A = $175 * 3/5 = 105$

33.33% of the total number of items sold by shop E = $300 * 1/3 = 100$

The required ratio = $105 : 100 = 21 : 20$

Hence, the correct answer is option B.

3. Questions

Shop A:

The total number of items sold by shop A = 200

The number of chairs sold by shop A = $(200 + 150)/2 = 175$

The number of tables sold by shop A = $200 - 175 = 25$

Shop B:

The total number of items sold by shop B = 350

The number of chairs sold by shop B = $(350 + 100)/2 = 225$

The number of tables sold by shop B = $350 - 225 = 125$

Shop C:

The total number of items sold by shop C = 250

The number of chairs sold by shop C = $(250 + 200)/2 = 225$

The number of tables sold by shop C = $250 - 225 = 25$

Shop D:

The total number of items sold by shop D = 150

The number of chairs sold by shop D = $(150 + 100)/2 = 125$

The number of tables sold by shop D = $150 - 125 = 25$

Shop E:

The total number of items sold by shop E = 300

The number of chairs sold by shop E = $(300 + 250)/2 = 275$

The number of tables sold by shop E = $300 - 275 = 25$

Shops	The number of chairs sold	The number of tables sold	The total number of items sold
A	175	25	200
B	225	125	350
C	225	25	250
D	125	25	150
E	275	25	300

Answer: C

The total number of tables sold by shops B and E together = $125 + 25 = 150$

The total number of items sold by shop D = 150

The required percentage = $150 / 150 * 100 = 100\%$

Hence, the correct answer is option C.

4. Questions

Shop A:

The total number of items sold by shop A = 200

The number of chairs sold by shop A = $(200 + 150)/2 = 175$

The number of tables sold by shop A = $200 - 175 = 25$

Shop B:

The total number of items sold by shop B = 350

The number of chairs sold by shop B = $(350 + 100)/2 = 225$

The number of tables sold by shop B = $350 - 225 = 125$

Shop C:

The total number of items sold by shop C = 250

The number of chairs sold by shop C = $(250 + 200)/2 = 225$

The number of tables sold by shop C = $250 - 225 = 25$

Shop D:

The total number of items sold by shop D = 150

The number of chairs sold by shop D = $(150 + 100)/2 = 125$

The number of tables sold by shop D = $150 - 125 = 25$

Shop E:

The total number of items sold by shop E = 300

The number of chairs sold by shop E = $(300 + 250)/2 = 275$

The number of tables sold by shop E = $300 - 275 = 25$

Shops	The number of chairs sold	The number of tables sold	The total number of items sold
A	175	25	200
B	225	125	350
C	225	25	250
D	125	25	150
E	275	25	300

Answer: A

The total number of chairs sold by shops A, C and E together = $175 + 225 + 275 = 675$

The total number of tables sold by shops B, C and D together = $125 + 25 + 25 = 175$

The required difference = $675 - 175 = 500$

Hence, the correct answer is option A.

5. Questions

Shop A:

The total number of items sold by shop A = 200

The number of chairs sold by shop A = $(200 + 150)/2 = 175$

The number of tables sold by shop A = $200 - 175 = 25$

Shop B:

The total number of items sold by shop B = 350

The number of chairs sold by shop B = $(350 + 100)/2 = 225$

The number of tables sold by shop B = $350 - 225 = 125$

Shop C:

The total number of items sold by shop C = 250

The number of chairs sold by shop C = $(250 + 200)/2 = 225$

The number of tables sold by shop C = $250 - 225 = 25$

Shop D:

The total number of items sold by shop D = 150

The number of chairs sold by shop D = $(150 + 100)/2 = 125$

The number of tables sold by shop D = $150 - 125 = 25$

Shop E:

The total number of items sold by shop E = 300

The number of chairs sold by shop E = $(300 + 250)/2 = 275$

The number of tables sold by shop E = $300 - 275 = 25$

Shops	The number of chairs sold	The number of tables sold	The total number of items sold
A	175	25	200
B	225	125	350
C	225	25	250
D	125	25	150
E	275	25	300

Answer: E

The average of the total number of items sold by shops B and C = $(350 + 250)/2 = 300$

$$3p = 300$$

$$p = 100$$

Hence, the correct answer is option E.

6. Questions

Shop A:

The total number of items sold by shop A = 200

The number of chairs sold by shop A = $(200 + 150)/2 = 175$

The number of tables sold by shop A = $200 - 175 = 25$

Shop B:

The total number of items sold by shop B = 350

The number of chairs sold by shop B = $(350 + 100)/2 = 225$

The number of tables sold by shop B = $350 - 225 = 125$

Shop C:

The total number of items sold by shop C = 250

The number of chairs sold by shop C = $(250 + 200)/2 = 225$

The number of tables sold by shop C = $250 - 225 = 25$

Shop D:

The total number of items sold by shop D = 150

The number of chairs sold by shop D = $(150 + 100)/2 = 125$

The number of tables sold by shop D = $150 - 125 = 25$

Shop E:

The total number of items sold by shop E = 300

The number of chairs sold by shop E = $(300 + 250)/2 = 275$

The number of tables sold by shop E = $300 - 275 = 25$

Shops	The number of chairs sold	The number of tables sold	The total number of items sold
A	175	25	200
B	225	125	350
C	225	25	250
D	125	25	150
E	275	25	300

Answer: A

The number of chairs sold by shop B = 225

The number of chairs sold by shop D = 125

The required difference = $225 - 125 = 100$

Hence, the correct answer is option A.

7. Questions

The length of the rectangular field = 25 m

Let the breadth of the rectangular field = b metre

The area of the rectangular field = length * breadth

So,

$$25 * b = 500$$

$$b = 20 \text{ m}$$

The perimeter of the rectangular field = $2(\text{length} + \text{breadth}) = 2(25 + 20) = 90 \text{ m}$

The perimeter of the square field = $4 * \text{side}$

The perimeter of the square field = $90 * \frac{2}{3} = 60 \text{ m}$

So,

$$4 * \text{side of the square field} = 60$$

The side of the square field = $60/4 = 15 \text{ m}$

So, $x = 15 \text{ m}$

Answer: A

The area of the square field = $15^2 = 225 \text{ m}^2$

The required cost = $225 * 12 = \text{Rs. } 2700$

Hence, the correct answer is option A.

8. Questions

The length of the rectangular field = 25 m

Let the breadth of the rectangular field = b metre

The area of the rectangular field = length * breadth

So,

$$25 * b = 500$$

$$b = 20 \text{ m}$$

The perimeter of the rectangular field = $2(\text{length} + \text{breadth}) = 2(25 + 20) = 90 \text{ m}$

The perimeter of the square field = $4 * \text{side}$

The perimeter of the square field = $90 * \frac{2}{3} = 60 \text{ m}$

So,

$$4 * \text{side of the square field} = 60$$

The side of the square field = $60/4 = 15 \text{ m}$

$$\text{So, } x = 15 \text{ m}$$

Answer: E

The area of the rectangular field = $25 * 20 = 500 \text{ m}^2$

The area of the square = $15^2 = 225 \text{ m}^2$

The required difference = $500 - 225 = 275 \text{ m}^2$

Hence, the correct answer is option E.

9. Questions

Shop P:

The total number of cakes sold by shop P = 250

The number of chocolate cakes sold by shop P = $250 * 40\% = 100$

The number of vanilla cakes sold by shop P = $250 - 100 = 150$

Shop Q:

The total number of cakes sold by shop Q = 375

The number of chocolate cakes sold by shop Q = $375 * 66.66\% = 250$

The number of vanilla cakes sold by shop Q = $375 - 250 = 125$

Shop R:

The total number of cakes sold by shop R = 128

The number of chocolate cakes sold by shop R = $128 * 75\% = 96$

The number of vanilla cakes sold by shop R = $128 - 96 = 32$

Shop S:

The total number of cakes sold by shop S = 320

The number of chocolate cakes sold by shop S = $320 * 55\% = 176$

The number of vanilla cakes sold by shop S = $320 - 176 = 144$

Shop T:

The total number of cakes sold by shop T = 200

The number of chocolate cakes sold by shop T = $200 * 67\% = 134$

The number of vanilla cakes sold by shop T = $200 - 134 = 66$

Shops	The number of chocolate cakes sold	The number of vanilla cakes sold	The total number of cakes sold
P	100	150	250
Q	250	125	375
R	96	32	128
S	176	144	320
T	134	66	200

Answer: D

The total number of cakes sold by shop U = $320 * 120\% = 384$

The number of chocolate cakes sold by shop U = $384 * 5/8 = 240$

Hence, the correct answer is option D.

10. Questions

Shop P:

The total number of cakes sold by shop P = 250

The number of chocolate cakes sold by shop P = $250 * 40\% = 100$

The number of vanilla cakes sold by shop P = $250 - 100 = 150$

Shop Q:

The total number of cakes sold by shop Q = 375

The number of chocolate cakes sold by shop Q = $375 * 66.66\% = 250$

The number of vanilla cakes sold by shop Q = $375 - 250 = 125$

Shop R:

The total number of cakes sold by shop R = 128

The number of chocolate cakes sold by shop R = $128 * 75\% = 96$

The number of vanilla cakes sold by shop R = $128 - 96 = 32$

Shop S:

The total number of cakes sold by shop S = 320

The number of chocolate cakes sold by shop S = $320 * 55\% = 176$

The number of vanilla cakes sold by shop S = $320 - 176 = 144$

Shop T:

The total number of cakes sold by shop T = 200

The number of chocolate cakes sold by shop T = $200 * 67\% = 134$

The number of vanilla cakes sold by shop T = $200 - 134 = 66$

Shops	The number of chocolate cakes sold	The number of vanilla cakes sold	The total number of cakes sold
P	100	150	250
Q	250	125	375
R	96	32	128
S	176	144	320
T	134	66	200

Answer: A

The total number of vanilla cakes sold by shops T and R = $32 + 66 = 98$

The number of chocolate cakes sold by shop S = 176

The required difference = $176 - 98 = 78$

Hence, the correct answer is option A.

11. Questions

Shop P:

The total number of cakes sold by shop P = 250

The number of chocolate cakes sold by shop P = $250 * 40\% = 100$

The number of vanilla cakes sold by shop P = $250 - 100 = 150$

Shop Q:

The total number of cakes sold by shop Q = 375

The number of chocolate cakes sold by shop Q = $375 * 66.66\% = 250$

The number of vanilla cakes sold by shop Q = $375 - 250 = 125$

Shop R:

The total number of cakes sold by shop R = 128

The number of chocolate cakes sold by shop R = $128 * 75\% = 96$

The number of vanilla cakes sold by shop R = $128 - 96 = 32$

Shop S:

The total number of cakes sold by shop S = 320

The number of chocolate cakes sold by shop S = $320 * 55\% = 176$

The number of vanilla cakes sold by shop S = $320 - 176 = 144$

Shop T:

The total number of cakes sold by shop T = 200

The number of chocolate cakes sold by shop T = $200 * 67\% = 134$

The number of vanilla cakes sold by shop T = $200 - 134 = 66$

Shops	The number of chocolate cakes sold	The number of vanilla cakes sold	The total number of cakes sold
P	100	150	250
Q	250	125	375
R	96	32	128
S	176	144	320
T	134	66	200

Answer: C

The number of chocolate cakes sold by shop P = 100

The number of vanilla cakes sold by shop Q = 125

The required percentage = $(125 - 100) / 125 * 100 = 20\%$

Hence, the correct answer is option C.

12. Questions

Shop P:

The total number of cakes sold by shop P = 250

The number of chocolate cakes sold by shop P = $250 * 40\% = 100$

The number of vanilla cakes sold by shop P = $250 - 100 = 150$

Shop Q:

The total number of cakes sold by shop Q = 375

The number of chocolate cakes sold by shop Q = $375 * 66.66\% = 250$

The number of vanilla cakes sold by shop Q = $375 - 250 = 125$

Shop R:

The total number of cakes sold by shop R = 128

The number of chocolate cakes sold by shop R = $128 * 75\% = 96$

The number of vanilla cakes sold by shop R = $128 - 96 = 32$

Shop S:

The total number of cakes sold by shop S = 320

The number of chocolate cakes sold by shop S = $320 * 55\% = 176$

The number of vanilla cakes sold by shop S = $320 - 176 = 144$

Shop T:

The total number of cakes sold by shop T = 200

The number of chocolate cakes sold by shop T = $200 * 67\% = 134$

The number of vanilla cakes sold by shop T = $200 - 134 = 66$

Shops	The number of chocolate cakes sold	The number of vanilla cakes sold	The total number of cakes sold
P	100	150	250
Q	250	125	375
R	96	32	128
S	176	144	320
T	134	66	200

Answer: E

The total number of chocolate cakes sold by shops P, R and T together = $100 + 96 + 134 = 330$

The required average = $330/3 = 110$

Hence, the correct answer is option E.

13. Questions

Shop P:

The total number of cakes sold by shop P = 250

The number of chocolate cakes sold by shop P = $250 * 40\% = 100$

The number of vanilla cakes sold by shop P = $250 - 100 = 150$

Shop Q:

The total number of cakes sold by shop Q = 375

The number of chocolate cakes sold by shop Q = $375 * 66.66\% = 250$

The number of vanilla cakes sold by shop Q = $375 - 250 = 125$

Shop R:

The total number of cakes sold by shop R = 128

The number of chocolate cakes sold by shop R = $128 * 75\% = 96$

The number of vanilla cakes sold by shop R = $128 - 96 = 32$

Shop S:

The total number of cakes sold by shop S = 320

The number of chocolate cakes sold by shop S = $320 * 55\% = 176$

The number of vanilla cakes sold by shop S = $320 - 176 = 144$

Shop T:

The total number of cakes sold by shop T = 200

The number of chocolate cakes sold by shop T = $200 * 67\% = 134$

The number of vanilla cakes sold by shop T = $200 - 134 = 66$

Shops	The number of chocolate cakes sold	The number of vanilla cakes sold	The total number of cakes sold
P	100	150	250
Q	250	125	375
R	96	32	128
S	176	144	320
T	134	66	200

Answer: B

The number of vanilla cakes sold by shop P = 150

The number of vanilla cakes sold by shop S = 144

The required ratio = $150 : 144 = 25 : 24$

Hence, the correct answer is option B.

14. Questions

Answer: A

The initial quantity of milk = $60 * \frac{5}{6} = 50$ litres

The initial quantity of water = $60 * \frac{1}{6} = 10$ litres

The removed quantity of milk = $30 * \frac{5}{6} = 25$ litres

The removed quantity of water = $30 * \frac{1}{6} = 5$ litres

According to the question,

$$(50 - 25) / (10 - 5 + y) = 5 / 6$$

$$25 / (5 + y) = 5 / 6$$

$$5 + y = 30$$

$$y = 25$$

Hence, the correct answer is option A.

15. Questions**Answer: C**Let the present age of P = $5a$ The present age of Q = $4a$

According to the question,

$$(5a + 3) / (4a + 3) = 11 / 9$$

$$45a + 27 = 44a + 33$$

$$a = 6$$

The present age of Q = $4 * 6 = 24$ yearsThe present age of P = $5 * 6 = 30$ yearsThe present age of R = $30 - 8 = 22$ yearsThe required sum = $24 + 22 = 46$ years

Hence, the correct answer is option C.

16. Questions**Answer: C**The profit sharing ratio of A, B and C = $(15000 * 12) : (24000 * 12) : (18000 * 8) = 5 : 8 : 4$ Let the profit share of A = $5a$ The profit share of B = $8a$ The profit share of C = $4a$

According to the question,

$$(5a + 8a + 4a) = 68000$$

$$17a = 68000$$

$$a = 4000$$

The profit share of C = $4 * 4000 = \text{Rs. } 16000$

Hence, the correct answer is option C.

17. Questions**Answer: D**Let the speed of the boat A = $4a$ The speed of the boat B = $5a$ The downstream speed of the boat A = $150/5 = 30$ km/hr

So,

$$(4a + 6) = 30$$

$$4a = 24$$

$$a = 6$$

The speed of the boat B = $5 * 6 = 30$ km/hr

The upstream speed of the boat B = $30 - 6 = 24$ km/hr

The required answer = $96/24 = 4$ hours

Hence, the correct answer is option D.

18. Questions

Answer: E

Let the cost price of the article = $100a$

The marked price of the article = $100a * 160\% = 160a$

The selling price of the article = $160a * 75\% = 120a$

According to the question,

$$120a - 100a = 800$$

$$20a = 800$$

$$a = 40$$

The cost price of the article = $100 * 40 = \text{Rs. } 4000$

The marked price of the article = $160 * 40 = \text{Rs. } 6400$

The new selling price of the article = $6400 * 85\% = \text{Rs. } 5440$

The required profit = $5440 - 4000 = \text{Rs. } 1440$

Hence, the correct answer is option E.

19. Questions

Answer: D

The ratio between the efficiency of A and B = $3 : 2$

Let the efficiency of A = 3 units per day

The efficiency of B = 2 units per day

The total work = $3 * 48 = 144$ units

The work done by A and B = $(3 + 2) * 12 = 60$ units

The remaining work = $144 - 60 = 84$ units

The efficiency of C = $84/6 = 14$ units per day

The combined efficiency of B and C = $14 + 2 = 16$ units per day

The required answer = $144/16 = 9$ days

Hence, the correct answer is option D.

20. Questions

Answer: E

The simple interest = $PRT/100$

So,

$$24300 = 18000 * 3 * r\%$$

$$r\% = 45\%$$

The rate of compound interest = $5 + 45 = 50\%$

The effective rate of compound interest = $50 + 50 + 2500/100 = 125\%$

The compound interest = $18000 * 125\% = \text{Rs. } 22500$

Hence, the correct answer is option E.

21. Questions

Answer: D

$$5/9 * 729 + 345 - ? * 3 = 15^2$$

$$405 + 345 - ? * 3 = 225$$

$$750 - ? * 3 = 225$$

$$? * 3 = 525$$

$$? = 175$$

Hence, the correct answer is option D.

22. Questions

Answer: A

$$\sqrt{6400} * 5 \div 8 = ?$$

$$80 * 5 \div 8 = ?$$

$$50 = ?$$

Hence, the correct answer is option A.

23. Questions

Answer: E

$$845 \div 5 - 25 * 4 + 41 * 5 = ?$$

$$169 - 100 + 205 = ?$$

$$69 + 205 = ?$$

$$274 = ?$$

Hence, the correct answer is option E.

24. Questions

Answer: A

$$(? + 72 \div 12) * 30 = 240$$

$$(? + 6) = 8$$

$$? = 2$$

Hence, the correct answer is option A.

25. Questions

Answer: C

$$(8)^3 * (16)^2 \div (32)^2 = (2)^?$$

$$128 = 2^?$$

$$2^? = 2^7$$

$$? = 7$$

Hence, the correct answer is option C.

26. Questions

Answer: A

$$(5.02/7.98) \text{ of } 319.95 + 19.99\% \text{ of } 749.98 = ? * 4.99$$

$$(5/8) * 320 + 20\% \text{ of } 750 = ? * 5$$

$$200 + 150 = ? * 5$$

$$350 = ? * 5$$

$$? = 70$$

Hence, the correct answer is option A.

27. Questions

Answer: B

$$32.02 * 14.99 - 18.01 * 11.95 + 14.99\% \text{ of } 600.15 = ?$$

$$32 * 15 - 18 * 12 + 15\% \text{ of } 600 = ?$$

$$480 - 216 + 90 = ?$$

$$354 = ?$$

Hence, the correct answer is option B.

28. Questions**Answer: C**

$$(215.98)^{1/3} * (728.95)^{1/3} + 14.99 * 5.95 = ?^2$$

$$(216)^{1/3} * (729)^{1/3} + 15 * 6 = ?^2$$

$$6 * 9 + 90 = ?^2$$

$$54 + 90 = ?^2$$

$$144 = ?^2$$

$$? = 12$$

Hence, the correct answer is option C.

29. Questions**Answer: D**

$$59.95\% \text{ of } 1250.23 + \sqrt{4760.98} * 3.99 = ? + 145.99$$

$$60\% \text{ of } 1250 + \sqrt{4761} * 4 = ? + 146$$

$$750 + 69 * 4 = ? + 146$$

$$750 + 276 = ? + 146$$

$$1026 = ? + 146$$

$$? = 880$$

Hence, the correct answer is option D.

30. Questions**Answer: E**

$$119.98 \div 4.95 * 15.01 - 25.02\% \text{ of } 439.98 = ?$$

$$120 \div 5 * 15 - 25\% \text{ of } 440 = ?$$

$$360 - 110 = ?$$

$$250 = ?$$

Hence, the correct answer is option E.

31. Questions**Answer: D**

$$6 * 1 - 2 = 4$$

$$4 * 2 - 3 = 5$$

$$5 * 3 - 4 = 11$$

$$11 * 4 - 5 = 39$$

$$39 * 5 - 6 = 189$$

Missing number: 5

Hence, the correct answer is option D.

32. Questions

Answer: C

$$46 + 13 = 59$$

$$59 + 17 = 76$$

$$76 + 19 = 95$$

$$95 + 23 = 118$$

$$118 + 29 = 147$$

Missing number: 147

Hence, the correct answer is option C.

33. Questions

Answer: B

$$7 + 1^3 = 8$$

$$8 + 2^2 = 12$$

$$12 + 3^3 = 39$$

$$39 + 4^2 = 55$$

$$55 + 5^3 = 180$$

Missing number: 7

Hence, the correct answer is option B.

34. Questions

Answer: A

$$16 * 0.5 + 1 = 9$$

$$9 * 1 + 1 = 10$$

$$10 * 1.5 + 1 = 16$$

$$16 * 2 + 1 = 33$$

$$33 * 2.5 + 1 = 83.5$$

Missing number: 16

Hence, the correct answer is option A.

35. Questions

Answer: E

$$13 + 5 = 18$$

$$18 + 7 = 25$$

$$25 + 9 = 34$$

$$34 + 11 = 45$$

$$45 + 13 = 58$$

Missing number: 45

Hence, the correct answer is option E.

36. Questions

Answer: A

$$11 * 1 + 1 = 12$$

$$12 * 2 + 2 = 26$$

$$26 * 3 + 3 = 81$$

$$81 * 4 + 4 = 328$$

$$328 * 5 + 5 = 1645$$

Wrong number: 10

Correct number: 11

Hence, the correct answer is option A.

37. Questions

Answer: B

$$9 + 13 = 22$$

$$22 + 26 = 48$$

$$48 + 39 = 87$$

$$87 + 52 = \mathbf{139}$$

$$139 + 65 = 204$$

Wrong number: 140

Correct number: 139

Hence, the correct answer is option B.

38. Questions

Answer: C

$$18 + 4^2 = 34$$

$$34 + 6^2 = 70$$

$$70 + 8^2 = 134$$

$$134 + 10^2 = 234$$

$$234 + 12^2 = \mathbf{378}$$

Wrong number: 380

Correct number: 378

Hence, the correct answer is option C.

39. Questions

Answer: D

$$21 \quad \mathbf{30} \quad 41 \quad 56 \quad 77 \quad 106$$

$$+9 \quad +11 \quad +15 \quad +21 \quad +29$$

$$+2 \quad +4 \quad +6 \quad +8$$

Wrong number: 29

Correct number: 30

Hence, the correct answer is option D.

40. Questions

Answer: E

$$33 + 29 = 62$$

$$62 + 31 = 93$$

$$93 + 37 = 130$$

$$130 + 41 = 171$$

$$171 + 43 = \mathbf{214}$$

Wrong number: 215

Correct number: 214

Hence, the correct answer is option E.

41. Questions

Answer: C

$$\mathbf{D). x^2 - 28x + 195 = 0}$$

$$x^2 - 15x - 13x + 195 = 0$$

$$x(x - 15) - 13(x - 15) = 0$$

$$(x - 13)(x - 15) = 0$$

$$x = 13 \text{ or } 15$$

$$\text{II). } y^2 - 30y + 224 = 0$$

$$y^2 - 16y - 14y + 224 = 0$$

$$y(y - 16) - 14(y - 16) = 0$$

$$(y - 14)(y - 16) = 0$$

$$y = 14 \text{ or } 16$$

So, $x = y$ or relationship cannot be established.

Hence, the correct answer is option C.

42. Questions

Answer: E

$$\text{I). } x^2 - 18x + 77 = 0$$

$$x^2 - 11x - 7x + 77 = 0$$

$$x(x - 11) - 7(x - 11) = 0$$

$$(x - 7)(x - 11) = 0$$

$$x = 7 \text{ or } 11$$

$$\text{II). } y^2 - 24y + 143 = 0$$

$$y^2 - 11y - 13y + 143 = 0$$

$$y(y - 11) - 13(y - 11) = 0$$

$$(y - 13)(y - 11) = 0$$

$$y = 13 \text{ or } 11$$

So, $x \leq y$

Hence, the correct answer is option E.

43. Questions

Answer: A

$$\text{I). } x^2 + 17x + 72 = 0$$

$$x^2 + 9x + 8x + 72 = 0$$

$$x(x+9) + 8(x+9) = 0$$

$$(x+8)(x+9) = 0$$

$$x = -8 \text{ or } -9$$

$$\text{II). } y^2 + 21y + 110 = 0$$

$$y^2 + 11y + 10y + 110 = 0$$

$$y(y+11) + 10(y+11) = 0$$

$$(y+10)(y+11) = 0 \text{ ki}$$

$$y = -10 \text{ or } -11$$

So, $x > y$

Hence, the correct answer is option A.

44. Questions

Answer: D

$$\text{I). } x^2 + 19x + 88 = 0$$

$$x^2 + 11x + 8x + 88 = 0$$

$$x(x+11) + 8(x+11) = 0$$

$$(x+11)(x+8) = 0$$

$$x = -11 \text{ or } -8$$

$$\text{II). } y^2 + 11y + 30 = 0$$

$$y^2 + 6y + 5y + 30 = 0$$

$$y(y+6) + 5(y+6) = 0$$

$$(y+5)(y+6) = 0$$

$$y = -5 \text{ or } -6$$

So, $x < y$

Hence, the correct answer is option D.

45. Questions

Answer: B

$$\text{I). } x^2 - 12x + 35 = 0$$

$$x^2 - 7x - 5x + 35 = 0$$

$$x(x-7) - 5(x-7) = 0$$

$$(x - 5)(x - 7) = 0$$

$$x = 5 \text{ or } 7$$

$$\text{II). } y^2 - 9y + 20 = 0$$

$$y^2 - 5y - 4y + 20 = 0$$

$$y(y - 5) - 4(y - 5) = 0$$

$$(y - 4)(y - 5) = 0$$

$$y = 4 \text{ or } 5$$

$$\text{So, } x \geq y$$

Hence, the correct answer is option B.

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